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Biochemical methane potential from sewage sludge: Effect of an aerobic pretreatment and fly ash addition as source of trace elements

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ABSTRACT

The effect of aerobic pretreatment and fly ash addition on the production of methane from mixed sludge is studied. Three assays with pretreated and not pretreated mixed sludge in the presence of fly ash (concentrations of 0, 10, 25, 50, 250 and 500 mg/L) were run at mesophilic condition. It was found that the combined use of aerobic pretreatment and fly ash addition increases methane production up to 70% when the fly ash concentrations were lower than 50 mg/L, while concentrations higher than 250 mg/L cause up to 11% decrease of methane production. For the anaerobic treatment of mixed sludge without pretreatment, the fly ash improved methane generation at all the concentrations studied, with a maximum of 56%. The removal of volatile solids does not show an improvement compared to the separate use of an aerobic pre-treatment and fly ash addition. Therefore, the combined use of the aerobic pre-treatment and fly ash addition improves only the production of methane.

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1. Introduction

Nowadays the continuous production of large amounts of sewage sludge is becoming a worldwide environmental problem (Zhao et al., 2010). Thus, emphasis in sludge management has shifted from its disposal to the beneficial use of itself and its products.

Among sludge treatment technologies, anaerobic digestion (AD) is a well-established and proven method. It transforms organic matter into biogas (60–70 vol% of methane, CH_4), destroys most of the pathogens present in the sludge, limits possible odor problems associated with residual putrescible matter, and reduces the amount of final sludge solids for disposal (Appels et al., 2008). However, the application of AD to sewage sludge is often limited by very long retention times (20–30 days) and a low overall degradation efficiency of the organic dry solids (30–50%). These limiting factors are generally associated with the hydrolysis stage (Tiehm et al., 2001). In order to improve this step, several sewage sludge pretreatments have been proposed, which include biological (largely thermal phase anaerobic), thermal hydrolysis, mechanical (ultrasound, high pressure, and lysis), chemical with oxidation and alkali treatments (Ennouri et al., 2016; Neumann et al., 2016). Among the biological methods, micro-aerobic pretreatment has been proposed (Girotto et al., 2016; Mata-Alvarez et al., 2000; Montalvo et al., 2016). It was reported that the limited oxygen supply caused an increase of the hydrolysis rates in the case of complex organic matter in batch tests (Johansen and Bakke, 2006), an improvement in digester performance, and a reduction of volatile fatty acid concentration (Diaz et al., 2011). Furthermore, the micro-aerobic step can be directly performed in the same unit prior to anaerobic digestion. This pretreatment has been used successfully to treat sewage sludge at the laboratory scale (Montalvo et al., 2016), increasing methane generation as well as volatile solids degradation.

In addition to pretreatment challenges, existing trace elements are insufficient for anaerobic microorganisms when solid waste, such as organic fractions of municipal solid waste or sewage sludge, is used alone as a feedstock for anaerobic digestion (Moestedt et al., 2016). The literature has emphasized the importance of supplementing trace elements for maximum methanogenic activity (Choong et al., 2016; Romero-Guiza et al., 2016; Thanh et al., 2016). Although the importance of this parameter is clear, its control is limited mainly by the high cost of the trace salts used for this purpose. In this context, the use of a cheaper source of trace elements could help control this parameter without additional expenses (Huilinir et al., 2015). A source of trace metals is fly ash (FA), which has been applied to the anaerobic digestion of municipal solid waste (Lo et al., 2010; Romero-Guiza et al., 2016) and recently to sewage sludge coming from the pulp and paper industry (Huilinir et al., 2015). The inclusion of fly ash in both cases

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increases methane production and also reduces the concentration of volatile solids (VS). However, the use of fly ash in the anaerobic digestion of sewage sludge has not been presented yet. Therefore, an analysis of the effect of fly ash on the anaerobic digestion of sewage sludge in the presence of trace elements is necessary.

Even though the use of a pretreatment and the use of fly ash as trace elements source had shown positive effects separately, the use of a combination both in the same substrate has not been studied yet. Only Mansour et al. (2012) compared the effect of an aerobic pretreatment and addition of bottom ashes (as pre-treatment) separately on anaerobic digestion of municipal solid waste, focusing on the kinetic of the process, without using the aerobic pretreatment and the bottom ashes combined. On the other hand, Liu et al. (2015) studying the anaerobic digestion of corn stover indicated that NaOH pretreatment and the addition of trace elements (Fe, Co and Ni) improve the biodegradability of corn stover and enhance biomethane production, and the present study is the only one that has dealt with the combined effect of pretreatment and trace element supplementation. Thus, the aim of the present work was to determine the combined effect of an aerobic pretreatment and the addition of fly ash as trace element source on methane production and VS reduction in a batch system. To the best of our knowledge, the combined effect of an aerobic pretreatment and fly ash as trace element source have not been reported previously. The effect of fly ash as the sole source of trace elements was also determined.

2. Materials and methods

2.1. Experimental setup

A 1-L glass reactor operating with an effective volume of 900 mL was used for the aerobic pretreatment. In each pretreatment air was injected continuously into the bioreactors by mini-compressors (0–10 l per minute, lpm) and distributed by small diffusers. The air flow rate (AFR) was calibrated before the pretreatment assays using the volume variation method. The conditions used for pretreatment were 0.35 vvm (air volume per liquid volume per minute) for 48 h at 35 °C, according to Montalvo et al. (2016). After aerobic pretreatment, pretreated sludge was fed to several batch anaerobic digesters of 280 mL of total volume.

For the anaerobic process, every experimental run considered the installation of twelve 280-mL anaerobic mini-digesters with effective operating volumes of 250 mL each. Eight of them were meant for measuring parameters in the liquid phase (chemical oxygen demand (COD) and volatile solids, and four were used to measure methane by liquid displacement according to Huilinir et al. (2015). The mini-digesters operated in discontinuous mode for 43 days, at which time the system reached a steady state. The digesters were stirred manually once or twice per day before reading the volume displaced by methane.

The volume of the digesters was completed with distilled water, they were stoppered with rubber stoppers and sealed with white silicone to ensure anaerobiosis, and they were covered with aluminum foil to prevent the growth of photosynthetic organisms. All the runs were carried out in duplicate.

2.2. Inocula, substrate and experimental design

The inoculum was obtained from an anaerobic reactor of the La Farfana Water Treatment Plant of Aguas Andinas in Santiago, Chile. The inoculum had 0.59 ± 0.01 g VS/g of total solids (TS), 2.2 ± 0.42 g of soluble COD/L, and a pH of 7.23 ± 0.12. The substrate was mixed sludge (a mixture of primary (60%) and secondary (40%) sludge) also from the La Farfana Water Treatment Plant. The substrate presented a ratio of 0.76 ± 0.01 g VS/g TS, 5.87 ± 0.87 g/L of soluble COD, and a pH of 5.69 ± 0.32. The main trace elements found in the mixed sludge were: Cd (1.2 mg/kg), Zn (3.19 mg/kg), Fe (1.80 mg/kg), Al (1.58 mg/kg), Zn (3.19 mg/kg), Cr (1.80 mg/kg), Cu (2.06 mg/kg), Ni (2.24 mg/kg), Mn (2.51 mg/kg), Ba (2.09 mg/kg), Co (0.8 mg/kg), B (0.74 mg/kg), Pb (2.09 mg/kg) and Se (2.77 mg/kg).

The sewage sludge was refrigerated at 4 °C for no more than 3 days until its use.

The fly ash was obtained from a thermoelectric power plant, with a particle diameter between 0.12 and 0.2 mm. The ash was taken from electrostatic precipitators used to collect particulate matter generated by the combustion of bituminous coal, placed before the gaseous effluents leave the plant. The main trace elements found in them were Fe (4.24 g/kg), Al (1.37 g/kg), Zn (5.8 mg/kg), Cr (3.93 mg/kg), Cu (2.32 mg/kg), Ni (3.85 mg/kg), Mn (37.7 mg/kg), V (26.4 mg/kg), Ba (64 mg/kg), Co (0.9 mg/kg), and B (25.4 mg/kg). Since the fly ash used in all the experiments came from the same sample obtained *in situ* from the thermoelectric power plant, the mass of trace elements used in all the experiments was proportional to the mass reported here.

In each anaerobic treatment, the substrate/inoculum ratio (F/I) was 1.5 g VS substrate/g VS inoculum according to Tomei et al. (2008). For each assay two weekly measurements of soluble and total COD and suspended VS were made in duplicate. The temperature in the container was kept at 35 °C using three automatically controlled aquarium heaters. Three assays were made in order to see the effect of fly ash and aerobic pretreatment, both separately and combined, as shown in Table 1. The first and second assays were ran on pretreated sludge (P) and different fly ash (FA) concentrations. The third assay was only meant to see the effect of fly ash on the anaerobic digestion.

2.3. Chemical analyses

The following parameters were determined: soluble COD and VS. COD was measured by a colorimetric method according to APHA (2012) and volatile solids (VS) were also measured according to APHA (2012).

Methane production was measured by volumetric displacement, connecting inverted Falcon tubes containing 3% w/w NaOH solution to remove CO_2 and H_2S as main impurities from the biogas, displacing only the methane volume. The resulting biogas travels through the flexible tubing to the Falcon tube, where it comes in contact with the NaOH solution, forming Na_2CO_3 and Na_2S and bubbling only methane. The methane collected in the

Table 1
Experimental conditions used in the anaerobic digestion assays.

Assay 1	Assay 2	Assay 3
Not pretreated, No FA* (NP-NFA1)	Not pretreated, No FA (NP-NFA2)	Not pretreated, No FA (NP-NFA3)
Pretreated, No FA (P-NFA1)	Pretreated, No FA (P-NFA2)	Pretreated, No FA (P-NFA3)
–	Pretreated, 10 mg/L FA (P-10)	Not pretreated, 10 mg/L FA (NP-10)
–	Pretreated, 25 mg/L FA (P-25)	Not pretreated, 25 mg/L FA (NP-25)
–	Pretreated, 50 mg FA/L (P-50)	Not pretreated, 50 mg/L FA (NP-50)
Pretreated, 250 mg/L FA (P-250)	–	Not pretreated, 250 mg/L FA (NP-250)
Pretreated, 500 mg/L FA (P-250)	–	Not pretreated, 500 mg/L FA (NP-500)

* FA: Fly Ash.

Falcon tube is sucked with a syringe and released in a flame to detect the presence of methane (Huilinir et al., 2015).

2.4. Statistical analyses

Statistical analyses were made with the STATGRAPHICS Centurion® software. In order to compare the data of each assay separately, *t*-test and the least significant difference (LSD) Fisher test were used, both with a 95% confidence interval.

The removal percentage was obtained from the following equations:

Volatile solids removal

$$\%R_{VS} = \left[\frac{VS_i - VS_f}{VS_i}\right] \cdot 100 \quad (1.1)$$

where VS_i represents the VS concentration on the initial day of the assay and VS_f represents the VS concentration at the end of the assay (day 43).

Since the real substrate of anaerobic microorganisms is the soluble COD, its removal is important in order to observe the effect of fly ash addition and aerobic pre-treatment on the process. Therefore, the soluble COD removal was calculated as:

$$\%R_{COD_S} = \left[\frac{COD_{S,i} - COD_{S_f}}{COD_{S_i}}\right] \cdot 100 \quad (1.2)$$

where $COD_{S,i}$ represents the soluble COD concentration on the initial day of the assay and $COD_{S,f}$ represents the soluble COD concentration at the end of assay (day 43).

3. Results and discussion

3.1. Effect on methane production

Fig. 1 shows the methane accumulated for the three assays. In Fig. 1 and Table 2 it is seen that pretreatment improved methane production compared with the system without pretreatment and without FA. For the first assay (Fig. 1a) the aerobic pretreatment increased methane production from 263 to 340 mL/$g_{initial\ VS}$ (29%); the second assay (Fig. 1b) showed a 30% increase (from 171 to 221 mL/$g_{initial\ VS}$), and in assay three (Fig. 1c) there is also an increase of up to 47% (from 195 to 287 mL/$g_{initialVS}$). This last value (47%) was higher than those obtained in assays 2 and 3, and it can be attributed to the different effect of the pretreatment on the composition and biodegradability of the substrate used in each anaerobic assay. The pre-aeration process, used here as an extra step where the sludge was aerated uninterruptedly during 48 h, improved the solubility of organic matter, making the rapidly biodegradable organic matter available to the fermentative and methanogenic biomass (Montalvo et al., 2016). The increase of the methanogenic activity can also be attributed to an improvement in the growth of facultative anaerobic microorganisms, which can keep a low redox potential, providing the best conditions for the growth of strict anaerobes (Mshandete et al., 2005). It is important to mention that the oxygen concentrations measured in the pre-aeration step were nil (not detected by sensors) under the aeration condition. Thus, the condition of "micro-aeration", i.e., oxygen concentration lower than 1 L O_2/L feed (Krayzelova et al., 2015) was accomplished and also showed that oxygen does not affect the anaerobic process after an aerobic pre-treatment (Montalvo et al., 2016).

The improvement in the methane generation agrees with the results presented by Montalvo et al. (2016), who reported an increase of up to 200% in accumulated methane when the sewage sludge was pretreated aerobically. Fu et al. (2015) also reported better performance of anaerobic digestion when the substrate (corn straw) was subjected to microaerobic pretreatment, increasing biogas production. Mansour et al. (2012) using pre-aeration for fresh municipal solid waste, found a significant advantage of the aerobic pretreatment over the control, with a much better performance of methanogenesis (higher methane production). The same trend was reported by Xu et al. (2014), who found an increase (between 5% and 18%) in methane generation in reactors with food waste pretreated aerobically. Thus, it is shown that pre-aeration can be used as pretreatment to increase methane yield in a batch system also for sewage sludge.

Fig. 1 also shows the effect of fly ash on the process. According to the results, the fly ash has different effects on methane production depending on whether the sewage sludge was subjected to pretreatment or not. When the sewage sludge was not pretreated (Fig. 1c), the effect of fly ash was always positive, with an increase in methane production varying from 15% at a fly ash concentration of 500 mg/L to 56% at a fly ash concentration of 50 mg/L, with the best performance between 10 and 50 mg/L (Table 2). In the range of 10–250 mg/L of fly ash, the accumulated methane does not show statistically significant differences ($p > 0.05$). A different situation was found for sewage sludge pretreated aerobically. Fig. 1a and Table 2 show decreasing methane production with respect to the NP-NFA1 assay at high fly ash concentrations, and decreasing methane production of 4% and 11% at fly ash concentrations of 250 and 500 mg/L, respectively. At lower fly ash concentrations (Fig. 1b) there is between 30 and 70% increase in methane production compared to the raw process (NP-NFA2), with the highest methane production achieved at 25 mg/L of FA (Table 2). It must be mentioned that statistically significant differences were determined in all the experiments with pretreatment compared to the assay of the pretreated sludge but without fly ash.

In order to see the effect of the fly ash on the assays separated from the effect of pretreatment (assays 2 and 3), methane production in the presence of FA was compared with the experiment without FA but with pretreatment (P-NFA); for the process without pretreatment the comparison was made with respect to the system without pretreatment and without fly ash (NP-NFA). The results are shown in Fig. 2, where it is seen that the FA in the aerobically pretreated system causes a strong inhibition of methane production, decreasing the methane accumulated when the fly ash concentrations are higher than 50 mg/L, with the highest decreasing around 30%. On the contrary, for the process without pretreatment, the effects of all the fly ash concentrations were positive, showing in all cases statistically significant differences ($p < 0.05$) in methane production with respect to the NP-NFA assay.

The positive effect of FA on the process obtained in this study agrees with other studies and can be related to the trace element concentrations in the FA. It is known that trace elements are necessary for a good anaerobic digestion performance. Zhang et al. (2011) indicated that the supplementation of trace elements in wastewater improved the performance of anaerobic digestion in terms of methane generation and volatile fatty acids decrease. Zhang and Jahng (2012) indicated that supplementation with trace elements keeps methane generation and pH constant in a continuous operation, showing that Fe was identified as the element that mostly affects this behavior. A similar situation was indicated by Facchin et al. (2013), who found that the mesophilic anaerobic digestion of food waste with two different inoculums improved methane production when the waste was supplemented with Se. The mechanism for accelerating and improving methane generation may be associated with the effect of trace elements on different types of enzymes, especially the metalloenzymes. According to the literature (Choong et al., 2016), the trace elements in the metalloenzyme are functioning to form the active site, the cofactor, and the structure itself. The general trace element requirement trend for metalloenzymes in methanogenesis can be simplified as

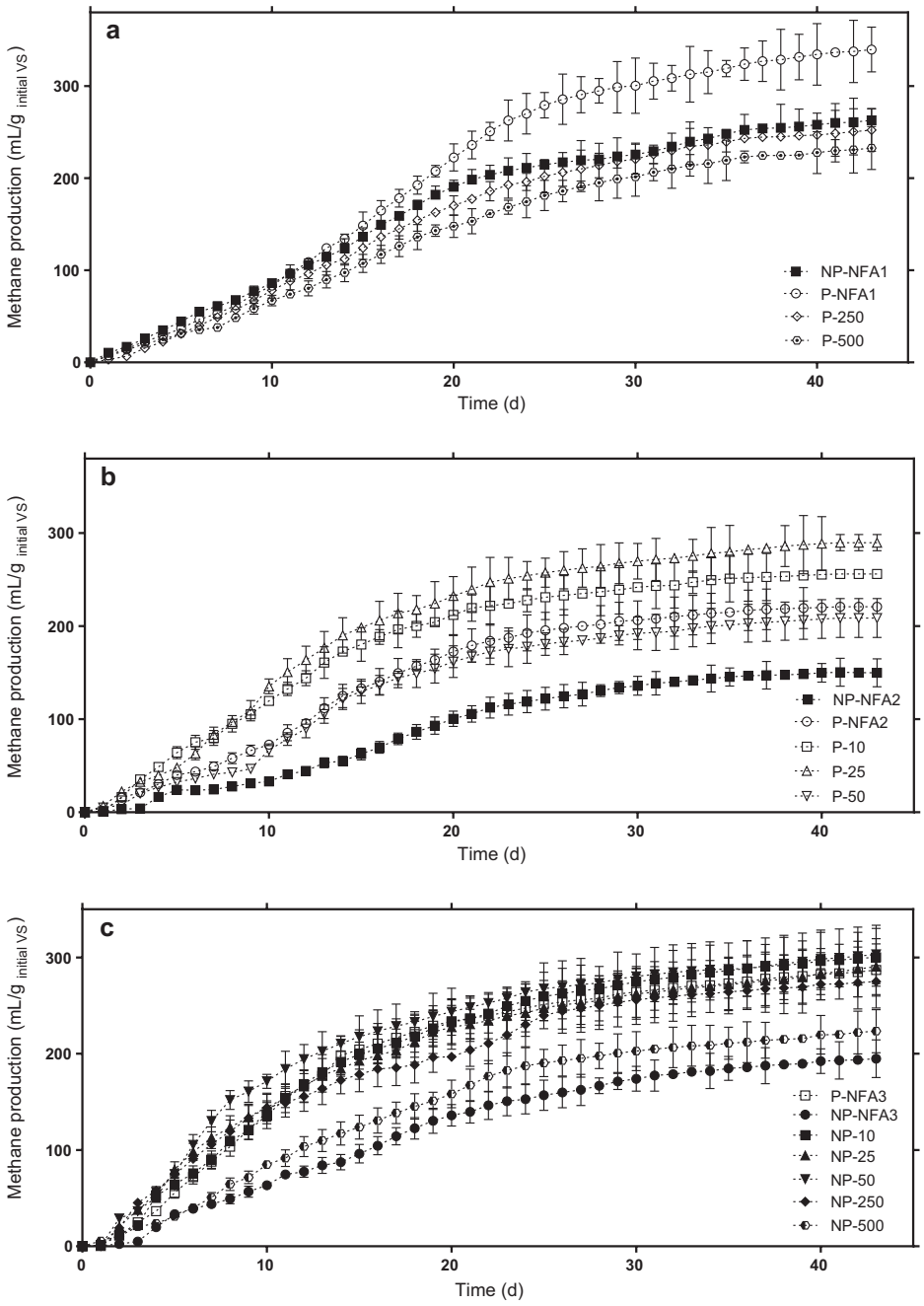


Fig. 1. Methane production in the assays with pretreatment and fly ash (a and b) and without pretreatment and fly ash (c).

Table 2
Methane accumulated at the end of assays for all the conditions studied.

	With aerobic pretreatment				Without aerobic pretreatment		
	Assay 1		Assay 2		Assay 3		
Experimental condition	Accumulated methane, mL $CH_4/g_{initialVS}$	Increase with respect to NP-NFA, %	Accumulated methane, mL $CH_4/g_{initialVS}$	Increase with respect to NP-NFA, %	Experimental condition	Accumulated methane, mL $CH_4/g_{initialVS}$	Increase with respect to NP-NFA, %
NP-NFA	263	–	171	–	NP-NFA	195	–
P-NFA	340	29	221	30	P-NFA	287	47
P-10	–	–	256	51	NP-10	300	54
P-25	–	–	290	70	NP-25	291	49
P-50	–	–	208	37	NP-50	303	56
P-250	252	−4	–	–	NP-250	275	41
P-500	233	−11	–	–	NP-500	223	15

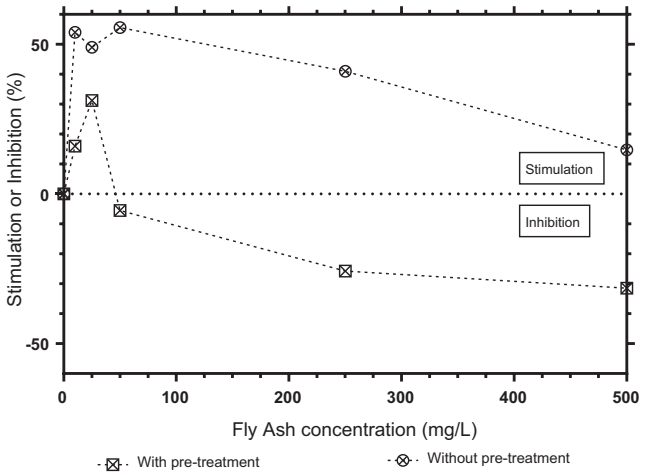


Fig. 2. Effect of fly ash concentrations on methane production for the process with and without pretreatment.

Fe ≫ Ni > Co > Zn = Mo/W (Choong et al., 2016; Glass and Orphan, 2012). This order is predicted without considering the substrate medium, which will shift the enzyme ratios, and hence alter the corresponding trace element requirement (Choong et al., 2016).

On the other hand, the negative effect seen with certain concentrations in this work has also been reported in the literature. Park et al. (2010), using simulated wastewater as substrate at 35 °C, showed that acetoclastic methanogens were not stimulated by nutrient supplementation at relatively low concentrations and a slight perturbation was observed when supplementation was at a relatively high concentration. Yue et al. (2007) studied the anaerobic digestion of cattail by rumen cultures in the presence of Cu(II), Cd(II) and Cr(VI), showing that low heavy metal concentrations (Cu(II) 2.4 mg/L, Cd(II) 1.6 mg/L and Cr(VI) 4.0 mg/L) promoted cattail degradation and methane production, but there is a severe inhibition of the cattail degradation at high heavy metal dosage. The chemical binding of heavy metals to the enzymes and subsequent disruption of the enzyme structure and function are the main causes of this toxic effect (Appels et al., 2008).

In order to explain the effect of fly ash on the process, it is necessary to see the different kinds of metals presents in the fly ash and how they may affect the process. Table 3 shows the elements and their effects reported in the literature, while Table 4 shows the concentration determined in each fly ash concentration used. According to this information, it can be said that Mn and Fe are the only trace elements whose concentrations are in the range found in the literature and reported as stimulatory and only when the fly ash concentrations are very high. Therefore, it is not possible to indicate which trace elements are affecting the process positively or negatively. However, because trace elements (mainly heavy metals) are present as mixtures, possible specific antagonistic and synergistic effects can explain the fly ash effect (Chen et al., 2014). Synergism means the enhanced toxic effect of one metal in the presence of small amounts of another metal, while antagonism refers to the effect of one metal which alleviates the toxic effect of another metal (Chen et al., 2014). For example, Oleszkiewicz and Sharma (1988) showed that aluminum silicate is an antagonist of the toxicity of other heavy metals. Earlier, Babich and Stotzky (1983) showed that nickel was synergistic in Ni–Cu, Ni–Mo–Co, and Ni–Hg systems, and antagonistic in Ni–Zn and Ni–Cd systems (Chen et al., 2014). Thus, it is the mixture of trace elements present in the FA and their interaction, and not the effect of only one trace element, that can explain the effect of fly ash in this study.

Table 3
Stimulating and inhibiting concentration ranges of trace elements present in the FA reported in the literature.

Metal	Stimulating concentration range (mg/L)	Inhibitory concentration range (mg/L)	References
Cr	0.01 < Cr < 15	27 < Cr < 2500	Romero-Guiza et al. (2016)
	4 < Cr < 15		Mao et al. (2015)
		200–250 (Cr^{+6})	Appels et al., 2008
		180–240 (Cr^{+3})	
		27	Altas (2009)
Mn	<0.027		Romero-Guiza et al. (2016)
Co	0.03 < Co < 19		Romero-Guiza et al. (2016)
	0.029 < Co < 5		Mao et al. (2015)
Ni	0.03 < Ni < 27	35 < Ni < 1600	Romero-Guiza et al. (2016)
	0.029 < Ni < 27		Mao et al. (2015)
		30	Appels et al. (2008)
		35	Appels et al. (2008)
Cu	0.03 < Cu < 2.4	12.5 < Cu < 350	Romero-Guiza et al. (2016)
		50–70	Appels et al. (2008)
Zn	0.03 < Zn < 2	7.5 < Zn < 1500	Mao et al. (2015), Romero-Guiza et al. (2016)
		1	Appels et al. (2008)
		7.5	Altas (2009)
Fe	<0.3		Romero-Guiza et al. (2016)
	1000–5000, in dry base		Mao et al. (2015)
Cd	<1.6	36 < Cd < 3400	Romero-Guiza et al. (2016)
		36	Altas (2009)
Pb	<0.2	67.2 < Pb < 8000	Romero-Guiza et al. (2016)
Se	<0.04		Romero-Guiza et al. (2016)
	0–10 mg/kg TS		Mao et al. (2015)
Mo	<0.05		Romero-Guiza et al. (2016)
	0.044–100		Mao et al. (2015)

According to the results obtained in the assays, the combination of an aerobic pretreatment and fly ash supplementation improved methane generation in comparison with a process only pretreated or only supplemented with fly ash, at least in a certain range of fly ash concentrations. This situation can be seen clearly in Fig. 2 and Table 2, where the combination of pretreatment and the use of 25 mg/L of fly ash increased methane production by up to 70%, while the use of pretreatment alone could increase methane generation by 48% and the use of only fly ash (assay 3) increased methane generation by up to 56%. However, the combination also decreases methane production when the fly ash concentrations are greater than 100 mg/L, as shown in Fig. 2. From these two effect it is possible to infer that: (i) Anaerobic digestion with pretreated sewage sludge required lower doses of trace elements, which was necessary only for the biological steps (acidogenesis, acetogenesis, and methanogenesis), and (ii) in raw sludge (without pretreatment), fly ash can influence both the biological steps and the hydrolytic step. This point can be seen in the greater VS removal obtained for the experiments without aerobic pretreatment, where the VS removal in the presence of fly ash was always greater than that of the system without fly ash, with the exception of the system with a fly ash concentration of 25 mg/L. The effect of trace elements on the hydrolytic step was mentioned by Zhang et al. (2014), who indicated that the use of Fe powder or scrap in sewage sludge promotes the hydrolysis-acidification of complex substances, and therefore provided more organic matter

Table 4
Trace element concentrations found in anaerobic test at different ash concentrations. In bold numbers the results that fall in the range given in the literature.

Element	mg/L				
	10 mg	25 mg	50 mg	250 mg	500 mg
Cd	0.00001	0.00003	0.00005	0.00026	0.00051
Zn	0.00006	0.00006	0.00029	0.00145	0.00290
Cr	0.00004	0.00004	0.00020	0.00098	0.00197
Cu	0.00002	0.00002	0.00012	0.00058	0.00116
Ni	0.00004	0.00004	0.00019	0.00096	0.00193
Pb	0.00002	0.00002	0.00010	0.00052	0.00105
Se	0.00003	0.00003	0.00014	0.00069	0.00139
Mn	0.00038	0.00038	0.00188	0.00942	**0.01883**
Co	0.00001	0.00001	0.00004	0.00022	0.00043
Mo	0.00001	0.00001	0.00004	0.00022	0.00045
Fe	0.04240	0.106	0.21199	**1.05996**	**2.11992**

for methanogenesis (Romero-Guiza et al., 2016). Liu et al. (2012) also indicated that the use of FeO on the acidogenic reactor of the two-stage sewage sludge digester improved COD removal, decreasing the volatile fatty acids in effluents, and improved the hydrolysis and fermentation. Finally, Meng et al. (2013) indicated that FeO addition to an acidogenic reactor enhanced the enzymatic activity 2–34 times. Therefore, the negative effect of fly ash on the anaerobic digestion at concentrations higher than 50 mg/L when the sewage sludge was pretreated could be explained as an excess of trace elements that was not used in the hydrolytic step.

According to these results, the combination of an aerobic pretreatment and fly ash as a source of trace elements improved methane production compared to the effect of fly ash and pretreatment separately, when fly ash concentrations are lower than 50 mg/L. Without pretreatment, the fly ash improved methane generation at all the concentrations studied (10–500 mg/L), with the best results in the range of 10–50 mg/L.

3.2. Effect of fly ash and aerobic pretreatment on volatile solids and soluble COD removal

Fig. 3 shows the concentrations of volatile solids (VS) and soluble COD in all the assays, while Fig. 4 shows the different removal percentages obtained. For the VS (Fig. 3a and b), it is seen that VS concentration decreases in both cases (with and without pretreatment), with the greatest decrease occurring during the first 15 days. This behavior indicates that the hydrolytic and acidogenic steps are predominant in the first days. According to the statistical analysis, there were no significant differences ($p > 0.05$) in VS concentrations between processes at different fly ash concentrations (Fig. 3a and b). Nevertheless, the trends of Fig. 3a and b showed that the process had a higher hydrolysis rate in the system without aerobic pretreatment in the first 15 days (Fig. 3b). Indeed, the average hydrolysis rate in the system with pretreatment in the first 15 days (determined on the slope observed on linear regression) was -423.89 ± 77.55 mg VS/L d, 15% lower than the average hydrolysis rate observed in the system without pre-treatment (501.43 ± 56.26 mg VS/L d). This situation indicates that the aerobic pretreatment degraded a part of the VS, decreasing the hydrolysis rate in the anaerobic step.

As shown in Fig. 4a, for the case with pretreatment the highest VS removal was around 33% and was obtained in the assay without fly ash (P-NFA), while the lowest VS removal occurred at fly ash concentrations of 25 and 50 mg/L, with 25% of VS removal. This behavior can be explained by the less biodegradable (hydrolysable) VS available in the system with pre-treated mixed sludge, in conjunction with the more highly soluble COD concentrations available for methanogenesis. This availability allows the methanogenic biomass to generate higher methane volumes without the need to hydrolyse more VS, which had been already degraded (at least a portion) in the aerobic pre-treatment. This situation can be observed too in the degradation of soluble COD in the first 15 days (Fig. 3c), where the decrease of soluble COD was more pronounced in the reactors with fly ash addition. Finally, the greater soluble COD removal obtained in the experiments with pre-treated sludge and fly ash addition in concentrations lower than 50 mg/L (Fig. 4b) also indicated that higher methane volumes are related to higher soluble COD removal. This also agreed with the higher methane production in the first 15 days (See Supplementary data) for the experiments with 10 and 25 mg/L of fly ash.

On the other hand, at higher fly ash concentrations (P-50, P-250 and P-500) there is lower methane production compared to the process with pretreatment and without fly ash (P-NFA), and therefore VS degradation was also slower.

Finally, there were significant differences ($p < 0.05$) in VS removal between processes with and without pretreatment with fly ash concentrations of 10, 25 and 50 mg/L.

It is important to note that the pre-treatment has an effect on the VS content. The original VS content in the mixed sludge was 0.76 ± 0.01 g_{VS}/g_{TS}, while after the pre-treatment it decreased to 0.73 ± 0.01 g_{VS}/g_{TS}, a value that indicated that the pre-treatment decreases the VS content in the mixed sludge. The decrease of the VS content in mixed sludge due to aerobic pre-treatment was analyzed in Montalvo et al. (2016), who indicated that an air flow rate of 0.3 vvm produced a 36% increase of the VSS hydrolysis rate compared to the system without pre-treatment. Therefore, the aerobic pre-treatment produces a faster hydrolysis rate of VS and also decreases the VS content available for anaerobic digestion.

The VS removal obtained in the pretreated sewage sludge was lower than that reported elsewhere. Fu et al. (2016) obtained 50% VS removal for a thermophilic anaerobic digestion of corn straw, while Montalvo et al. (2016) reported that the removal was as high as 68%. The differences can be related to the type of substrate and also the different initial VS concentrations used in each case.

For the case without pretreatment (Fig. 4a), the tendency of VS removal was different, because the highest VS removal was achieved in the presence of fly ash. In this case, the process without pretreatment and without fly ash (NP-NFA) has a low VS removal (around 30%), showing that, as mentioned earlier, the addition of fly ash probably improved the hydrolytic step. The highest VS removal (around 40%) was obtained with 50 mg/L of fly ash, and the lowest removal (28%) with 25 mg/L of fly ash, which also showed one of the highest methane productions (Table 2). These results agree with the results obtained by Huilinir et al. (2015), but it results do not agree with those reported by Lo et al. (2009), who studied the use of fly ash on the anaerobic digestion of municipal solid waste. They indicated that VS removal was lower in the system with fly ash because under these conditions the microbial activity was higher, as reflected in higher methane

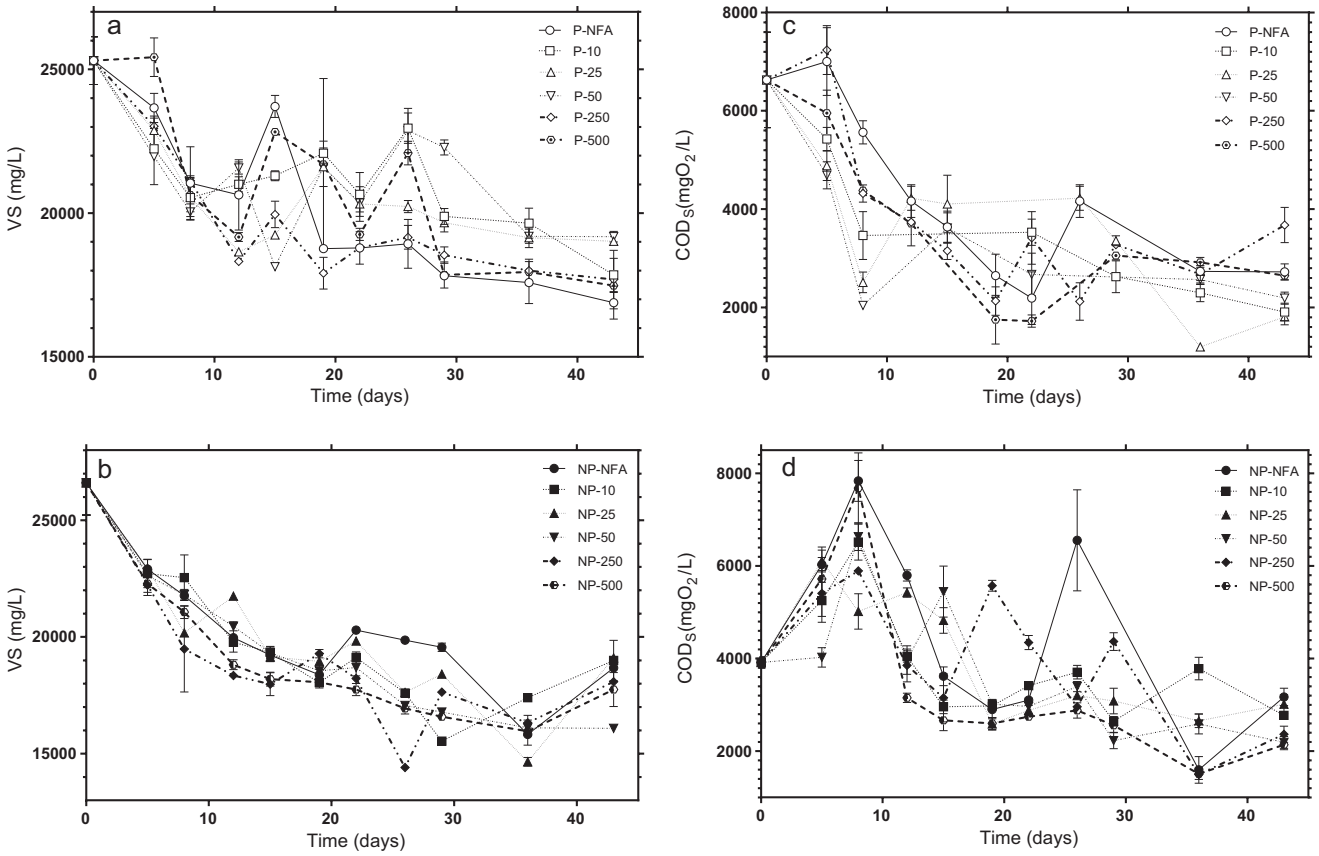


Fig. 3. Volatile solids and soluble COD in the assays. (a) VS in assays with pretreatment; (b) VS in assays without pretreatment; (c) soluble COD in assays with pretreatment; (d) soluble COD in assays without pretreatment.

yields. This behavior occurred partially in this work, because only at 25 mg/L of fly ash the VS removal was lower than that obtained without fly ash. Therefore, for anaerobic digestion of sewage sludge the use of fly ash without pretreatment improved VS removal, while for aerobically pretreated sewage sludge the fly ash does not have a positive effect on VS removal.

Regarding soluble COD, different concentration profiles were obtained with and without pretreatment. When the sewage sludge was pretreated (Fig. 3c), there was a clear decrease in all the assays, with the initial soluble COD concentration (around 6500 mg COD/L) higher than that in the anaerobic process without pretreatment, mainly because the aerobic pretreatment increased the hydrolysis efficiency. Indeed, the concentration of soluble COD before the aerobic pre-treatment was 5.87 ± 0.87 g/L, while after the pre-treatment it increased up to 10.4 ± 0.04 g/L. These high concentrations also explain the high removal of soluble COD achieved in these assays, which was greater than 40% (Fig. 4b). It is important to note that the soluble COD consumption rate (determined from the slopes found for the linear regressions) in the first 15 days of the experiments was higher in the system with fly ash up to 50 mg/L. For instance, the soluble COD consumption rate in the system without fly ash addition was −223.91 mg COD/L d, while at concentrations of 10, 25 and 50 mg/L the COD consumption rate was −362.96, −481.03 and −511.53 mg COD/L d, respectively. These values also agreed with the higher methane volume produced in these experiments (see Supplementary data). Therefore, the fly ash addition up to values of 50 mg/L increased the soluble COD consumption rate in the process.

The removal percentage of soluble COD (Fig. 4b) is greater at lower fly ash values compared to the process with pretreatment and without fly ash, while at high fly ash concentrations (P-250 and P-500) the soluble COD removal was lower than in the process without fly ash (P-NFA). This behavior is in good agreement with the tendency observed for methane production (Fig. 2), where it was seen that high fly ash concentrations decrease methane production.

When the sewage sludge was not pretreated, the initial soluble COD concentrations were low (close to 4000 mg/L) compared to those of the system with pretreatment, and it starts increasing on day 10, reaching values between 6000 to 8000 mg COD/L. This initial increase can be attributed to the hydrolytic step and has also been found in other studies (Huilinir et al., 2015; Montalvo et al., 2016). The increase of soluble COD concentration was more pronounced in the system without fly ash addition and with a fly ash concentration of 500 mg/L, with soluble COD generation rates of 483.33 and 459.91 mg COD/L d for the system without fly ash and 500 mg/L with fly ash, respectively (Fig. 3d). For the rest of the fly ash concentrations studied, the generation rate was lower, mainly because the fly ash addition could stimulate COD consumption under these conditions. After this initial increase, the soluble COD starts decreasing due to the action of methanogenic biomass, reaching values of around 3000 mg COD/L by the end of the assays. The removal percentage for the process without pretreatment was low, varying between 20 and 43% for all the assays. The soluble COD removal tendency also differs from that seen in the assays with pretreatment, showing a clear improvement in the removal when the added fly ash concentration was greater. This behavior also agrees with the methane production tendency observed in the process without pretreatment, showing that the fly ash for that process improves the microbial activity and therefore the COD degradation.

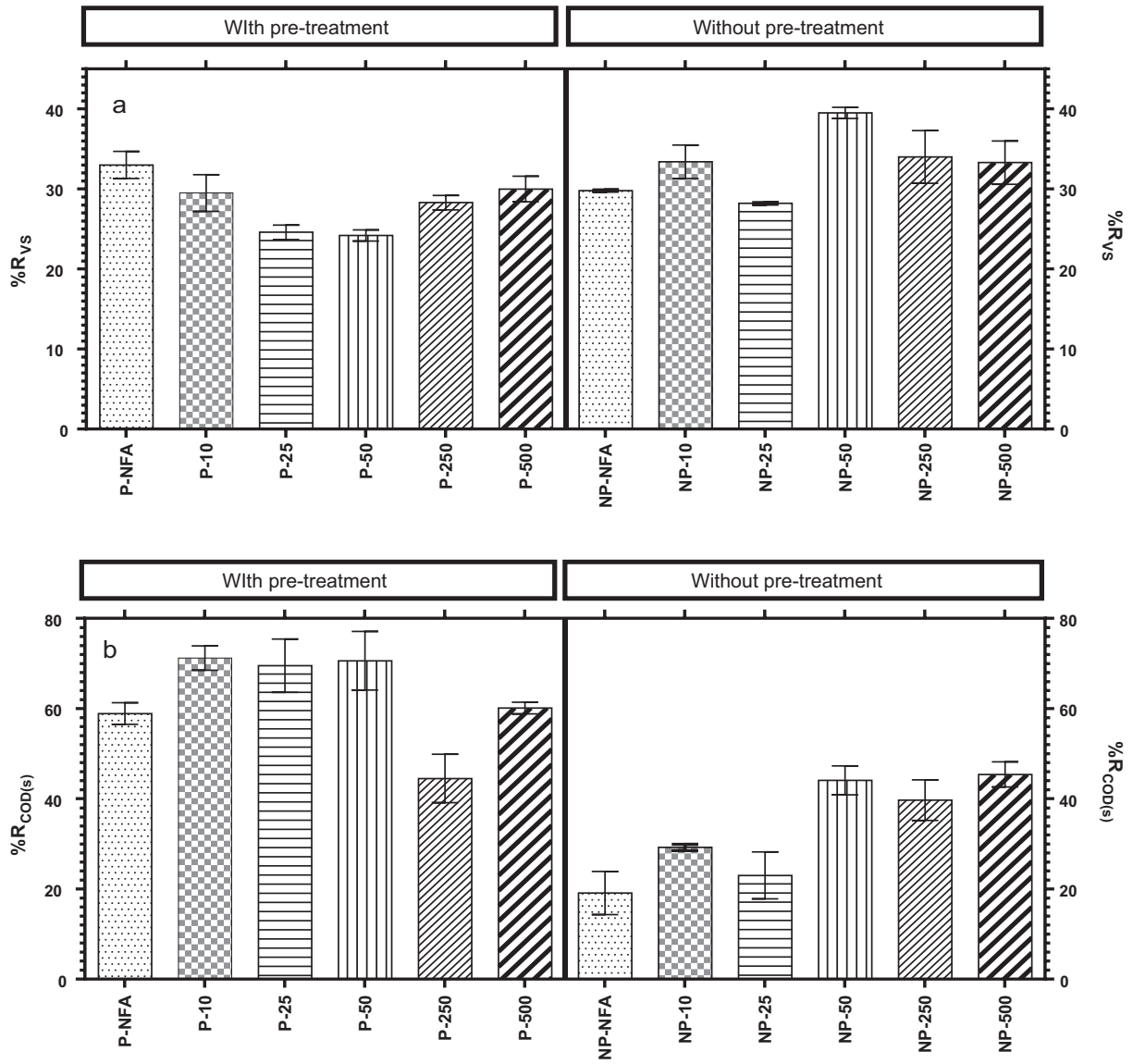


Fig. 4. Removal percentage of VS (a) and CODs (b).

Regarding the statistical analysis for soluble COD, it shows that there are significant differences between the system with and without pretreatment for all the fly ash concentrations studied ($p < 0.05$).

Thus, according to these results, it is possible to say that the combined use of aerobic pretreatment and fly ash as trace element supplier improved the removal of soluble COD, which was higher with fly ash concentrations between 10 and 50 mg/L.

It is known that the digested sewage sludge (digestate) can be recycled by application in agricultural land (Alvarenga et al., 2015; Brisolara and Qi, 2013); however, it can represent a potential risk to agricultural soils because of their potential contamination with heavy metals, organic contaminants and/or pathogens. Due to the application of fly ash as a source of trace elements that also contains several heavy metals in the anaerobic digestion, it is important to know if the digestate complies with current regulations. Chile has Decree-Law No. 4/2009 which indicates the maximum amount of some heavy metals in order to apply biosolids to soils. Therefore, the amount of some heavy metals in the digestate for one of the highest fly ash concentration used (250 mg/L) was measured. The results are shown in Table 5. It is seen that none of the heavy metal concentrations in the digestate was higher than the values recommended by the Chilean regulations for application to agricultural soils, but the amount of heavy metals in the digestate is higher than the amount found in the fly ash, showing that the addition of fly ash increased the total amount of heavy metals compared to the mixed sludge and the fly ash. Therefore, the application of fly ash in the anaerobic digestion of mixed sludge does not affect the feasibility of applying the digestate in agricultural soils.

Table 5
Amount of trace elements founded in digested mixed sludge (total concentration and dry base) after the process without aerobic pre-treatment and with addition of 250 mg/L of fly ash.

Element	Digested mixed sludge, mg/kg	Chilean Legislation mg/kg	Fly ash, mg/kg
Cd	1.34	8	N.D.
Zn	470.74	2000	5.6
Cu	155.69	1000	2.32
Ni	9.08	80	3.85
Pb	13.69	300	N.D.
Se	N.D.	50	N.D.

4. Conclusions

The combined use of aerobic pretreatment and fly ash improved methane production when fly ash concentrations were lower than 50 mg/L. Fly ash concentrations higher than 50 mg/L inhibited the methanogenic process, with lower methane production compared to the process with pretreatment and without fly ash. VS removal does not improve by the combined use of an aerobic pretreatment and fly ash. For the anaerobic treatment of sewage sludge without pretreatment, fly ash improved methane generation and VS removal at all the concentrations studied (10–500 mg/L).

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.wasman.2017.03.023.

References

Alvarenga, P., Mourinha, C., Farto, M., Santos, T., Palma, P., Sengo, J., Morais, M.C., Cunha-Queda, C., 2015. Sewage sludge, compost and other representative organic wastes as agricultural soil amendments: benefits versus limiting factors. Waste Manage. 40, 44–52.

APHA, AWWA, WPCF, 2012. Standard Methods for the Examination of Water and Wastewater, 22th ed. APHA Washington, DC.

Appels, L., Baeyens, J., Degreve, J., Dewil, R., 2008. Principles and potential of the anaerobic digestion of waste-activated sludge. Prog. Energy Combust. Sci. 34, 755–781.

Altas, L., 2009. Inhibitory effect of heavy metals on methane-producing anaerobic granular sludge. J. Hazard. Mater. 162, 1551–1556.

Babich, H., Stotzky, G., 1983. Synergism between nickel and copper in their toxicity to microbes: mediation by pH. Ecotoxicol. Environ. Saf. 7, 576–587.

Brisolara, K.F., Qi, Y.A., 2013. Biosolids and sludge management. Water Environ. Res. 85, 1283–1297.

Chen, J.L., Ortiz, R., Steele, T.W.J., Stuckey, D.C., 2014. Toxicants inhibiting anaerobic digestion: a review. Biotechnol. Adv. 32, 1523–1534.

Choong, Y.Y., Norli, I., Abdullah, A.Z., Yhaya, M.F., 2016. Impacts of trace element supplementation on the performance of anaerobic digestion process: a critical review. Bioresour. Technol. 209, 369–379.

Diaz, I., Donoso-Bravo, A., Fdz-Polanco, M., 2011. Effect of microaerobic conditions on the degradation kinetics of cellulose. Bioresour. Technol. 102, 10139–10142.

Ennouri, H., Miladi, B., Diaz, S.Z., Guelfo, L.A.F., Solera, R., Hamdi, M., Bouallagui, H., 2016. Effect of thermal pretreatment on the biogas production and microbial communities balance during anaerobic digestion of urban and industrial waste activated sludge. Bioresour. Technol. 214, 184–191.

Facchin, V., Cavinato, C., Fatone, F., Pavan, P., Cecchi, F., Bolzonella, D., 2013. Effect of trace element supplementation on the mesophilic anaerobic digestion of foodwaste in batch trials: the influence of inoculum origin. Biochem. Eng. J. 70, 71–77.

Fu, S.F., Shi, X.S., Xu, X.H., Wang, C.S., Wang, L., Dai, M., Guo, R.B., 2015. Secondary thermophilic microaerobic treatment in the anaerobic digestion of corn straw. Bioresour. Technol. 186, 321–324.

Fu, S.F., Wang, F., Shi, X.S., Guo, R.B., 2016. Impacts of microaeration on the anaerobic digestion of corn straw and the microbial community structure. Chem. Eng. J. 287, 523–528.

Girotto, F., Peng, W., Rafieenia, R., Cossu, R., 2016. Effect of aeration applied during different phases of anaerobic digestion. Waste Biomass Valoriz., 1–14

Glass, J.B., Orphan, V.J., 2012. Trace metal requirements for microbial enzymes involved in the production and consumption of methane and nitrous oxide. Front. Microbiol. 3, 61.

Huilinir, C., Montalvo, S., Guerrero, L., 2015. Biodegradability and methane production from secondary paper and pulp sludge: effect of fly ash and modeling. Water Sci. Technol. 72, 230–237.

Johansen, J.E., Bakke, R., 2006. Enhancing hydrolysis with microaeration. Water Sci. Technol. 53, 43–50.

Krayzelova, L., Bartacek, J., Diaz, I., Jeison, D., Volcke, E.I.P., Jenicek, P., 2015. Microaeration for hydrogen sulfide removal during anaerobic treatment: a review. Rev. Environ. Sci. Bio-Technol. 14, 703–725.

Liu, C., Yuan, H., Zou, D., Liu, Y., Zhu, B., Li, X., 2015. Improving biomethane production and mass bioconversion of corn stover anaerobic digestion by adding NaOH pretreatment and trace elements. Biomed. Res. Int.

Liu, Y.W., Zhang, Y.B., Quan, X., Li, Y., Zhao, Z.Q., Meng, X.S., Chen, S., 2012. Optimization of anaerobic acidogenesis by adding Fe-0 powder to enhance anaerobic wastewater treatment. Chem. Eng. J. 192, 179–185.

Lo, H.M., Kurniawan, T.A., Sillanpaa, M.E.T., Pai, T.Y., Chiang, C.F., Chao, K.P., Liu, M. H., Chuang, S.H., Banks, C.J., Wang, S.C., Lin, K.C., Lin, C.Y., Liu, W.F., Cheng, P.H., Chen, C.K., Chiu, H.Y., Wu, H.Y., 2010. Modeling biogas production from organic fraction of MSW co-digested with MSWI ashes in anaerobic bioreactors. Bioresour. Technol. 101, 6329–6335.

Lo, H.M., Liu, M.H., Pai, T.Y., Liu, W.F., Lin, C.Y., Wang, S.C., Banks, C.J., Hung, C.H., Chiang, C.F., Lin, K.C., Chen, P.H., Chen, J.K., Chiu, H.Y., Su, M.H., Kurniawan, T.A., Wu, K.C., Hsieh, C.Y., Hsu, H.S., 2009. Biostabilization assessment of MSW co-disposed with MSWI fly ash in anaerobic bioreactors. J. Hazard. Mater. 162, 1233–1242.

Mansour, A.A., Motte, A., Pallier, V., Feuillade-Cathalifaud, G., Ponthieux, A., 2012. Assessment of the aerobic preparation and bottom ash addition as pretreatment steps before landfilling: impact on methanogenesis kinetics and leachate parameters. Waste Manage. 32, 1864–1872.

Mao, C.L., Feng, Y.Z., Wang, X.J., Ren, G.X., 2015. Review on research achievements of biogas from anaerobic digestion. Renew. Sust. Energ. Rev. 45, 540–555.

Mata-Alvarez, J., Mace, S., Llabres, P., 2000. Anaerobic digestion of organic solid wastes. An overview of research achievements and perspectives. Bioresour. Technol. 74, 3–16.

Meng, X.S., Zhang, Y.B., Li, Q., Quan, X., 2013. Adding Fe-0 powder to enhance the anaerobic conversion of propionate to acetate. Biochem. Eng. J. 73, 80–85.

Moestedt, J., Nordell, E., Shakeri Yekta, S., Lundgren, J., Martí, M., Sundberg, C., Ejlertsson, J., Svensson, B.H., Björn, A., 2016. Effects of trace element addition on process stability during anaerobic co-digestion of OFMSW and slaughterhouse waste. Waste Manage. 47 (Part A), 11–20.

Montalvo, S., Huiliñir, C., Ojeda, F., Castillo, A., Lillo, L., Guerrero, L., 2016. Microaerobic pretreatment of sewage sludge: effect of air flow rate, pretreatment time and temperature on the aerobic process and methane generation. Int. Biodeterior. Biodegradation 110, 1–7.

Mshandete, A., Bjornsson, L., Kivaisi, A.K., Rubindamayugi, S.T., Mattiasson, B., 2005. Enhancement of anaerobic batch digestion of sisal pulp waste by mesophilic aerobic pre-treatment. Water Res. 39, 1569–1575.

Neumann, P., Pesante, S., Venegas, M., Vidal, G., 2016. Developments in pre-treatment methods to improve anaerobic digestion of sewage sludge. Rev. Environ. Sci. Bio-Technol. 15, 173–211.

Oleszkiewicz, J.A., Sharma, V.K., 1988. Role of Heavy Metals in Anaerobic Digestion, Heavy Metals in the Hydrological Cycle. Publications Division, Selper Ltd., London, pp. 583–592.

Park, C., Bega, A., Unlu, C., Chadderton, R.A., McKean, W.R., Kohl, P.M., Hunt, J.A., Keaney, J., Willis, J.L., Duran, M., 2010. Acetoclastic methanogens in an anaerobic digester could be susceptible to trace metal supplementation. Water Sci. Technol. 62, 2905–2911.

Romero-Guiza, M.S., Vila, J.J., Mata-Alvarez, J., Chimenos, J.M., Astals, S., 2016. The role of additives on anaerobic digestion: a review. Renew. Sust. Energy Rev. 58, 1486–1499.

Thanh, P.M., Ketheesan, B., Yan, Z., Stuckey, D., 2016. Trace metal speciation and bioavailability in anaerobic digestion: a review. Biotechnol. Adv. 34, 122–136.

Tiehm, A., Nickel, K., Zellhorn, M., Neis, U., 2001. Ultrasonic waste activated sludge disintegration for improving anaerobic stabilization. Water Res. 35, 2003–2009.

Tomei, M.C., Braguglia, C.M., Mininni, G., 2008. Anaerobic degradation kinetics of particulate organic matter in untreated and sonicated sewage sludge: role of the inoculum. Bioresour. Technol. 99, 6119–6126.

Xu, S.Y., Selvam, A., Wong, J.W.C., 2014. Optimization of micro-aeration intensity in acidogenic reactor of a two-phase anaerobic digester treating food waste. Waste Manage. 34, 363–369.

Yue, Z.B., Yu, H.Q., Wang, Z.L., 2007. Anaerobic digestion of cattail with rumen culture in the presence of heavy metals. Bioresour. Technol. 98, 781–786.

Zhang, L., Jahng, D., 2012. Long-term anaerobic digestion of food waste stabilized by trace elements. Waste Manage. 32, 1509–1515.

Zhang, L., Lee, Y.W., Jahng, D., 2011. Anaerobic co-digestion of food waste and piggery wastewater: focusing on the role of trace elements. Bioresour. Technol. 102, 5048–5059.

Zhang, Y.B., Feng, Y.H., Yu, Q.L., Xu, Z.B., Quan, X., 2014. Enhanced high-solids anaerobic digestion of waste activated sludge by the addition of scrap iron. Bioresour. Technol. 159, 297–304.

Zhao, L., Gu, W.M., He, P.J., Shao, L.M., 2010. Effect of air-flow rate and turning frequency on bio-drying of dewatered sludge. Water Res. 44, 6144–6152.